

Name: _____ Reg #: _____ Section: _____

National University of Computer and Emerging Sciences, Lahore Campus

Course: Artificial Intelligence

Program: BS(Computer Science)

Duration: 45 Minutes

Paper Date: 6-May-2026

Semester: Spring 2026

Course Code: AI2002

Section: BCS-6C

Total Marks: 20

Page(s): 2

Exam: Quiz 3

Q1) A regression model predicts house prices based on property features.

(3+2+2=8)

Input Features:

x1: House size (100 sq ft units)

x2: Number of rooms

For a house:

$$x_1 = 2, x_2 = 3$$

Hidden layer:

h₁: w₁₁ = 0.3, w₂₁ = 0.4, b₁ = 0.1

h₂: w₁₂ = 0.5, w₂₂ = -0.2, b₂ = -0.1

Activation: Linear

Output layer:

o: w₁ = 1.2, w₂ = 0.8, b = 0.5

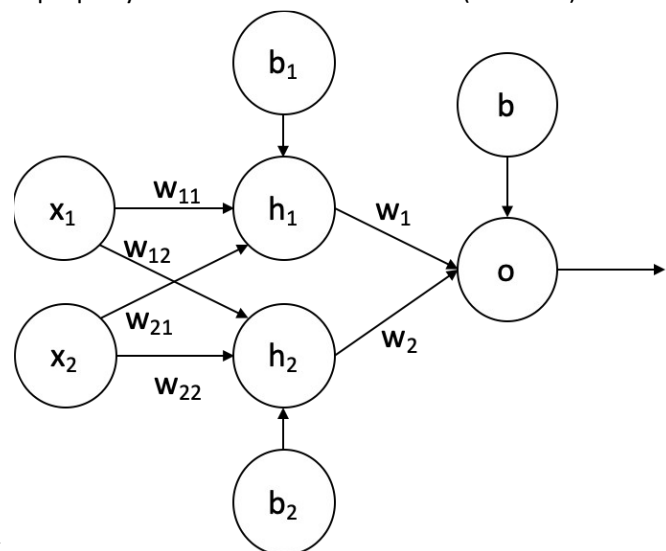
Activation: Linear

Target:

$$y = 10$$

Loss/Error: MSE = $\frac{1}{2}(y - \hat{y})^2$

Hint (Derivatives): derivative of a linear function is 1



(a) Forward pass (compute all neuron outputs)

Unit	Net Input	Output
h ₁	$(0.3)(2) + (0.4)(3) + 0.1 = 1.9$	1.9
h ₂	$(0.5)(2) + (-0.2)(3) - 0.1 = 0.3$	0.3
o	$(1.2)(1.9) + (0.8)(0.3) + 0.5 = 3.02$	3.02

(b) Compute gradients for output weights w₁, w₂

	Gradients (O-y)(1)(input)
$\partial E / \partial w_1$	$(3.02 - 10)(1)(1.9) = -13.262$
$\partial E / \partial w_2$	$(3.02 - 10)(1)(0.3) = -2.094$

(c) Update output weights and bias using learning rate $\eta = 0.1$

	New weights ($w_{old} - \eta \partial E / \partial w$) and bias ($b_{old} - \eta \partial E / \partial b$)
w ₁	$1.2 - (0.1)(-13.262) = 2.5262$
w ₂	$0.8 - (0.1)(-2.094) = 1.0094$
b	$0.5 - (0.1)(-6.98) = 1.198$

Q2) A data analyst is clustering customers based on two features: Income (in \$1000s) and Spending Score. (2+4+2=8)

Customer	Income (X)	Spending Score (Y)
C1	2	6
C2	3	5
C3	8	7
C4	9	6
C5	8	2
C6	9	3

The analyst applies **K-means with K = 2** and obtains the following clusters after convergence:

- **Cluster A:** C1, C2, C3, C4
- **Cluster B:** C5, C6

(a) Compute the **centroid of each cluster**.

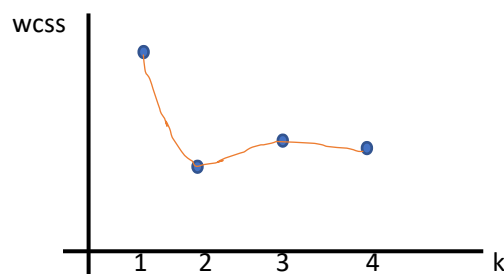
Centroid for Cluster A	Centroid for Cluster B
(5.5, 6)	(8.5, 2.5)

(b) Calculate the **Within-Cluster Sum of Squares (WCSS)** for each cluster using $(x-x_c)^2+(y-y_c)^2$

Cluster A	WCSS
C1	12.25
C2	7.25
C3	7.25
C4	12.25
Total	39.0

Cluster B	WCSS
C5	0.5
C6	0.5
Total	1

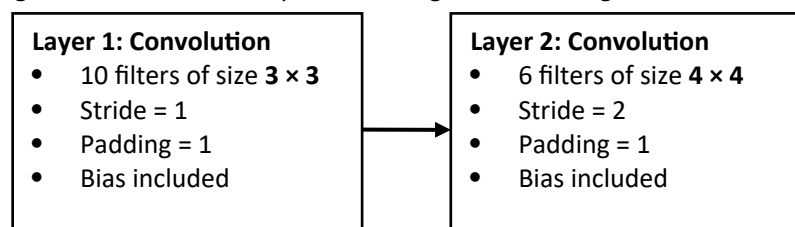
(c) For the given WCSS values for different K using the Elbow Method, determine the best value of K.



K	WCSS
1	110.0
2	39+1 = 40
3	45.0
4	44.5

Q3) An RGB image of size **80 × 80 × 3** is passed through the following CNN:

(4)



Calculate the following:

	Layer 1	Layer 2
Output Dimensions	80 × 80 × 10	40 × 40 × 6
Parameters	280	966